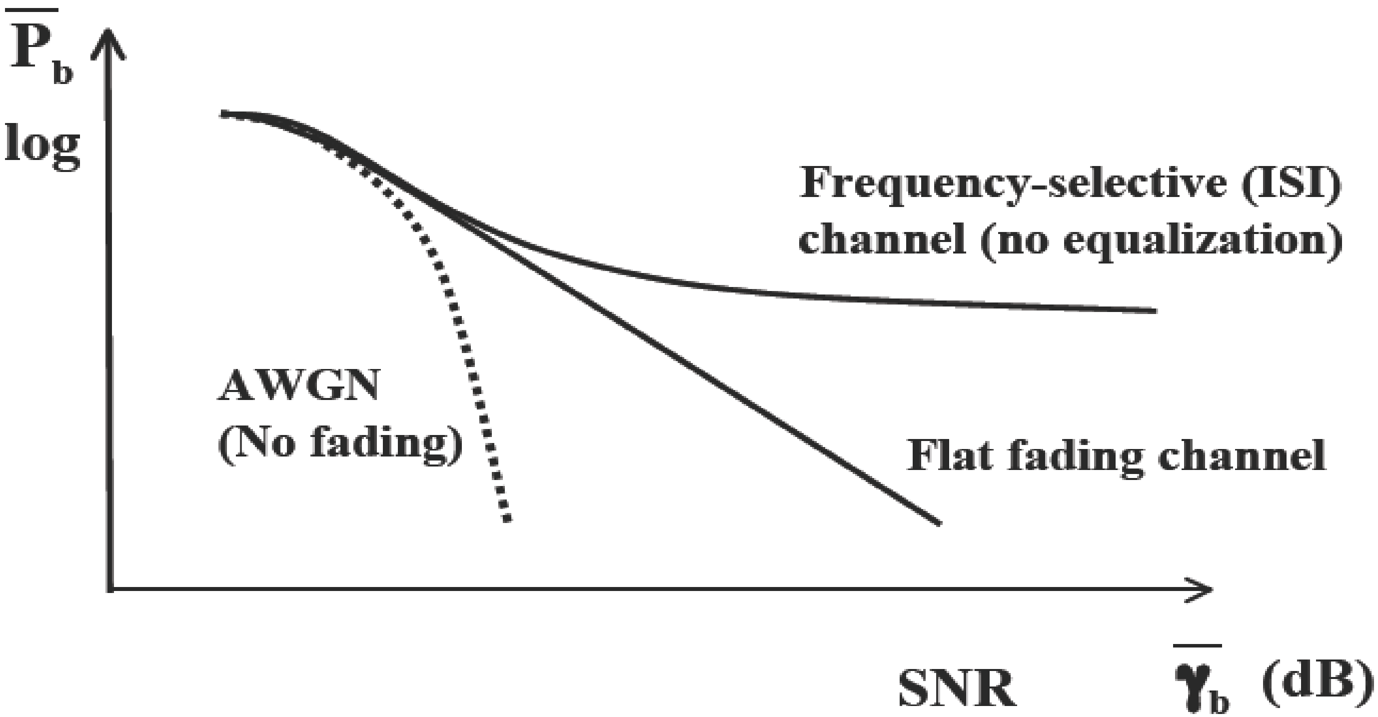
# Wireless Digital Communications

## Topic 8

## Multicarrier Modulation-OFDM

# Typical BER Versus SNR Curves




$$SINR = P_r/(N_0B + ISI) \qquad SNR = P_r/(N_0B)$$

$$P_r \uparrow \rightarrow \left\langle \begin{matrix} SNR \uparrow \\ ISI \uparrow \end{matrix} \right\rangle \rightarrow SINR \quad \text{remains} \quad \text{unchanged}$$

# Outline

- ISI Countermeasures
- Multicarrier Modulation
- Overlapping Subsystems
- Fading Across Subcarriers
- Discrete Implementation of MCM
- FFT Implementation of OFDM
- OFDM Challenges

# ISI Countermeasures

- Equalization
  - Signal processing at receiver to eliminate ISI, must balance ISI removal with noise enhancement
  - Can be very complex at high data rates, and performs poorly in fast-changing channels
  - Not that common in state-of-the-art wireless systems
- Multicarrier Modulation
  - Break data stream into lower-rate substreams modulated onto narrowband flat-fading subchannels
- Spread spectrum
  - Superimpose a fast (wideband) spreading sequence on top of data sequence, allows resolution for combining or attenuation of multipath components.

# Flat and Frequency-Selective Fading Channels

$$A_C(\Delta f) = E\left[C^*(f;t)C(f+\Delta f;t)\right] = \int_{-\infty}^{\infty} A_C(\tau)e^{-j2\pi\Delta f\tau}\,d\tau$$

**(distribution of average power in frequency)**

**= 0 implies uncorrelated (in frequency) fading**

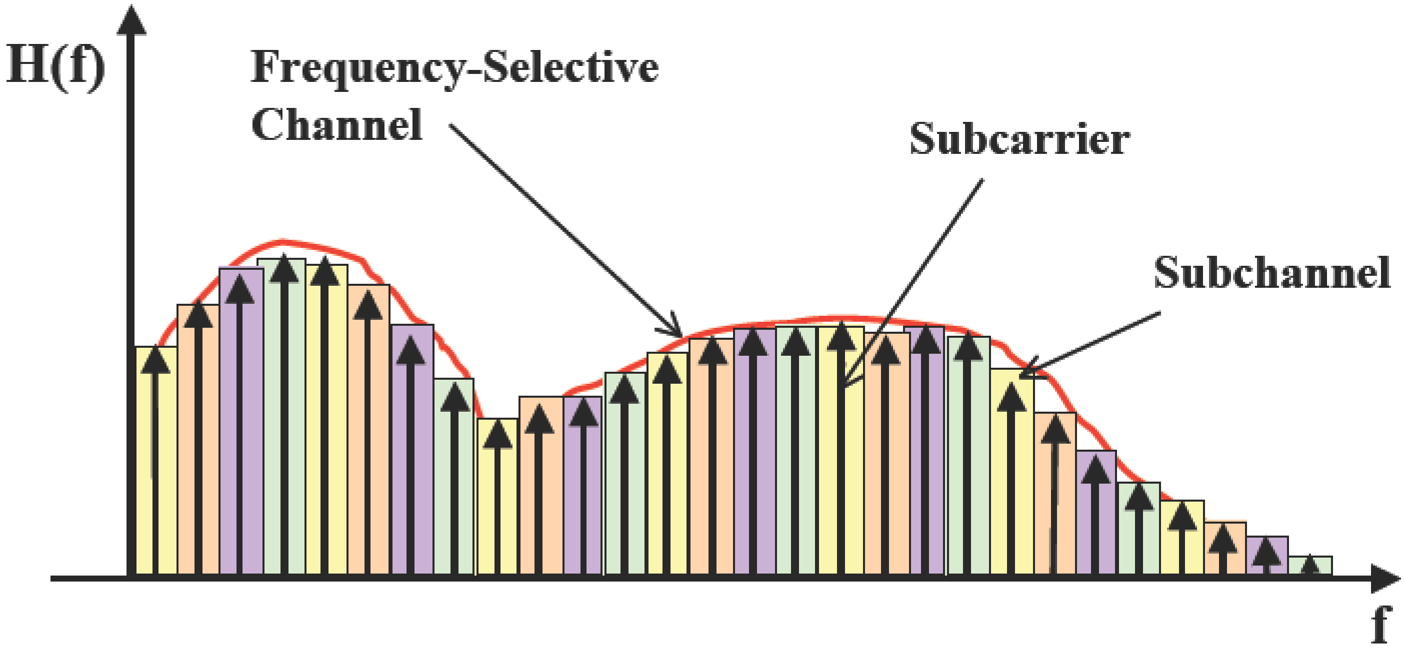
| | | | | | |
|---|---|---|---|---|---|
| $T_s$ | **= Symbol period** | $B$ | $= 1/T_s$ | **Signal BW** |
| $T_m$ | **= Delay spread** | $B_c$ | $= 1/T_m$ | **Coherence BW** |

$T_s \gg T_m \Rightarrow B \ll B_c \Rightarrow$ **flat fading (no ISI)**

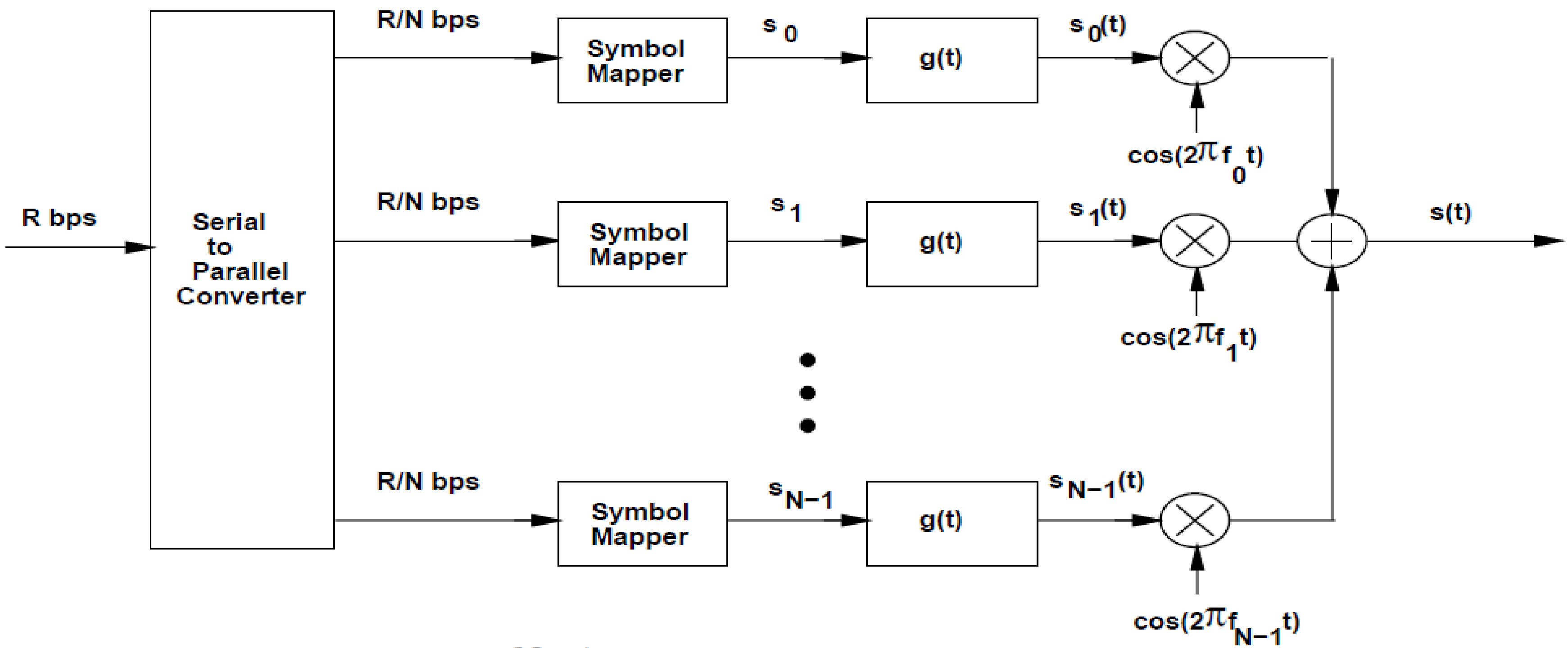
$T_s \ll T_m \Rightarrow B \gg B_c \Rightarrow$ **frequency-selective fading (severe ISI)**

# Multicarrier Modulation

Frequency-selective (ISI) channel is divided into many narrow parallel subchannels. Each subchannel is a narrow flat-fading Channel, leading to no ISI



# Multicarrier Transmitter for BPSK




$$s(t) = \sum_{i=0}^{N-1} s_i g(t) \cos(2\pi f_i t + \phi_i)$$

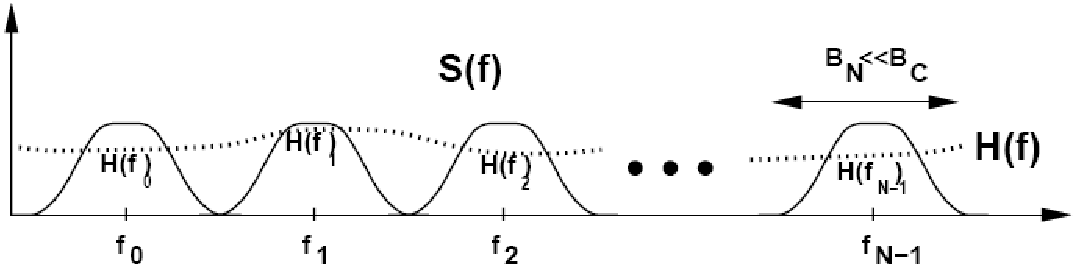
# Multicarrier Modulation Non-Overlapping Subchannels

**B :Total bandwidth , N: No. of Subchannels,**

**Bandwith of subchannel $B_N = B/N$, $T_N = (1+\beta)/B_N$,**

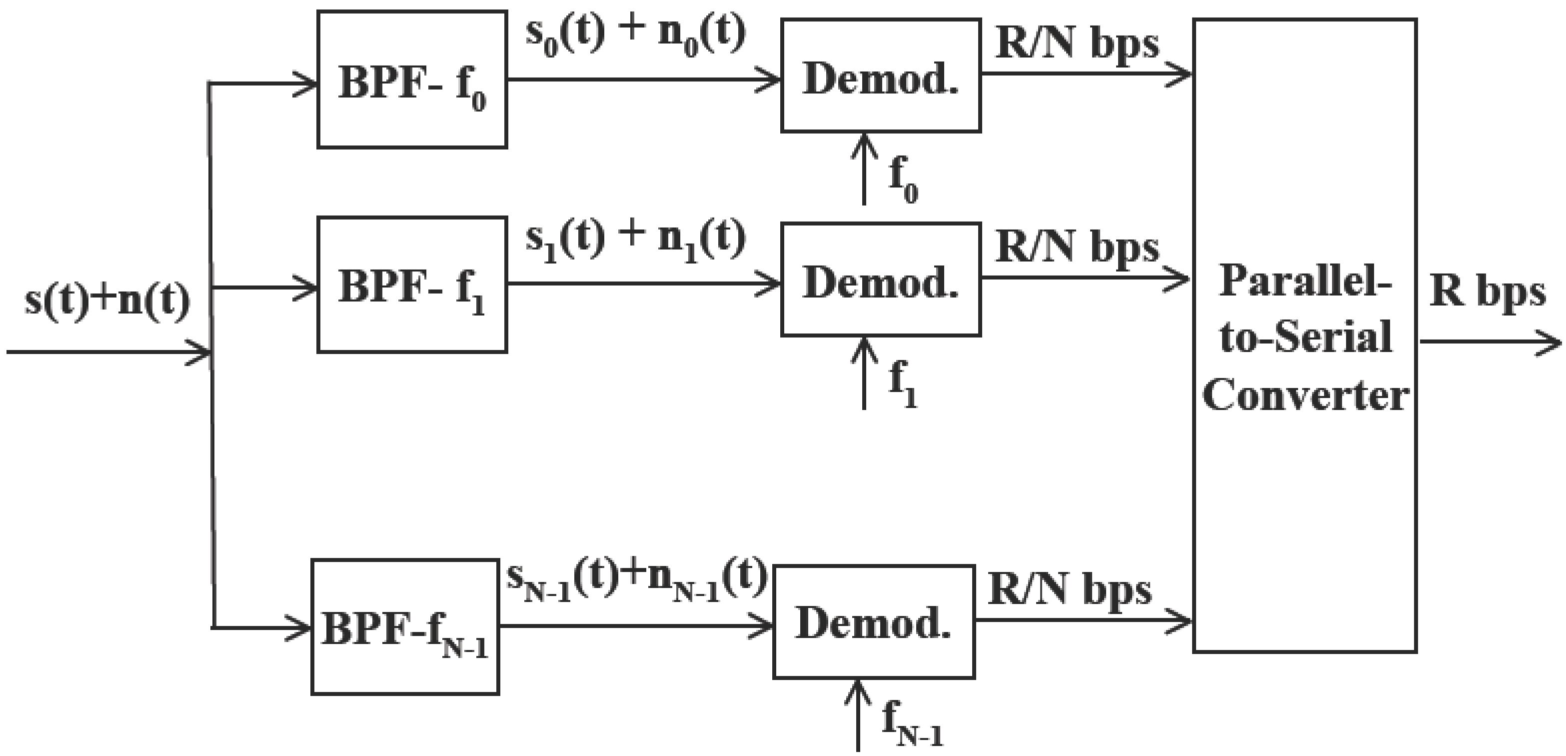
**Total Sym. Data Rate $R = N/T_N = NB_N/(1+\beta) = B/(1+\beta)$**

**$f_i = f_0 + iB_N$, $i = 0,1,\ldots,N-1$ subchannel frequencies**




*Transmitted Signal*

# Traditional Multicarrier Receiver



# Drawbacks of Traditional MCM

**In practice:**

(i) **Sharp BPF at receiver not practical**

(ii) **N independent modulator/demodulators needed**

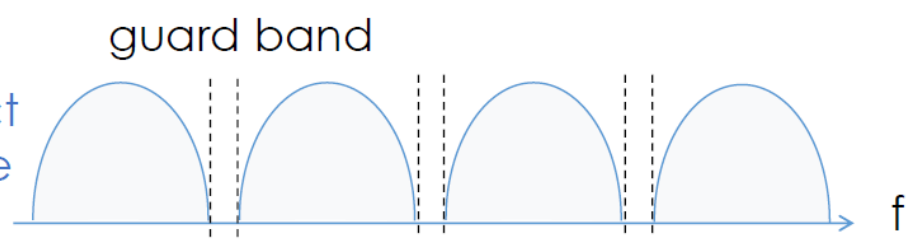
(iii) **Subchannel bandwidths are larger**

**Total BW:** $$B = \frac{N(1+\beta+\varepsilon)}{T_N} = (1+\beta+\varepsilon)R$$

$\beta / T_N$ **additional BW of raised cosine pulse shape**

$\varepsilon / T_N$ **additional BW of time-limited raised cosine**

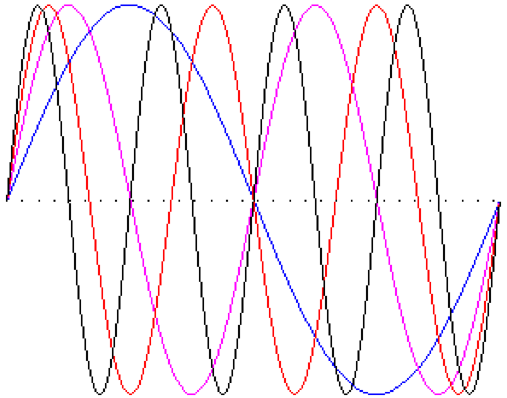
Guard bands protect leakage interference



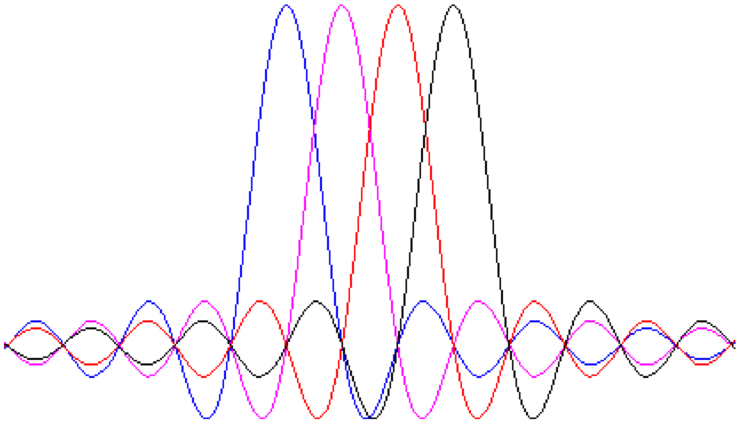
# MCM-Overlapping Subchannels (OFDM)

**Solution: Overlapping Subchannels (Sharp filtering is not required.)**

**B≈N/T*N*=R**



Time domain



Frequency domain

# Overlapping Subchannels Orthogonality

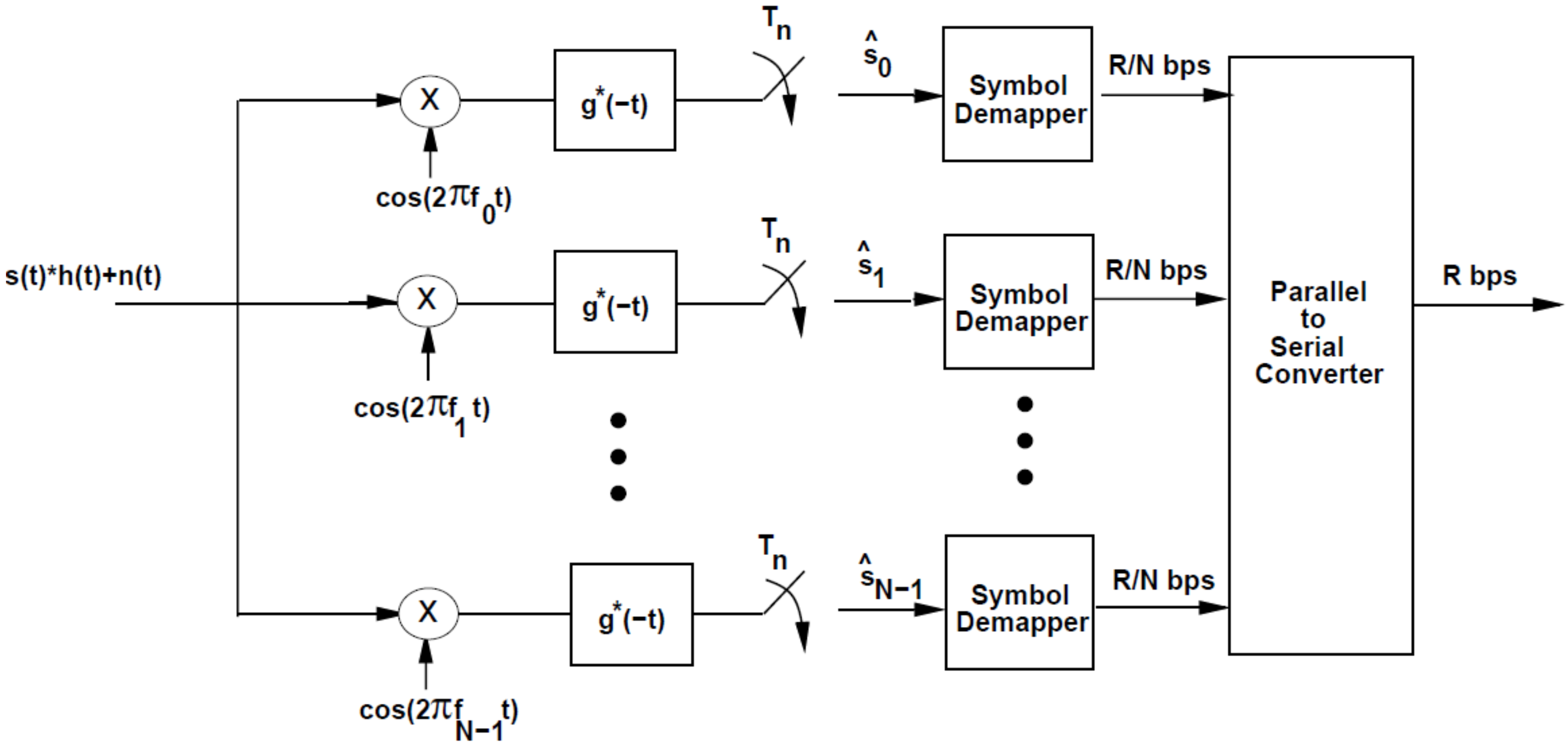
**subcarriers at $f_0 + i/T_N$ and $f_0 + j/T_N$ are orthogonal:**

$$\frac{1}{T_N}\int_0^{T_N} \cos\left(2\pi\left(f_0 + \frac{i}{T_N}\right)t + \phi_i\right)\cos\left(2\pi\left(f_0 + \frac{j}{T_N}\right)t + \phi_i\right)dt = \begin{matrix} 0.5 & i = j \\ 0 & i \neq j \end{matrix}$$

**Minimum subcarrier requirement: $1/T_N$**

**Receiver requires N subcarrier demodulators at $f_0 + i/T_N$ ($i = 0, 1, \ldots, N-1$). Subchannels are separated using orthogonality of subcarriers.**

# MCM-Overlapping Subchannels Receiver Structure for BPSK



# MCM-Overlapping Subchannels Signal Recovery

$$\hat{s}_i = \int_0^{T_N} \left( \sum_{j=0}^{N-1} s_j g(t) \cos(2\pi f_j t + \phi_j) \right) g(t) \cos(2\pi f_i t + \phi_i) dt$$

$$= \sum_{j=0}^{N-1} s_j \int_0^{T_N} g^2(t) \cos\left(2\pi\left(f_0 + \frac{j}{T_N}\right)t + \phi_i\right) \cos\left(2\pi\left(f_0 + \frac{i}{T_N}\right)t + \phi_i\right) dt$$

$$= \sum_{j=0}^{N-1} s_j \delta(j-i) = s_i$$

**Fading across subcarriers is a problem with MCM (both overlapping and non-overlapping)**

# Fading Across Subcarriers

- Leads to different BERS due to different channel gains $\alpha_i$ for different subcarrier bands.

- Compensation techniques
  - Frequency equalization
  - Precoding
  - Coding across subcarriers
  - Adaptive loading (power and rate)

# Subcarrier Fading-Compensations

- Frequency equalization:  Channel inversion at the receiver
  - Requires channel knowledge at receiver only
  - Impact of fading removed, but noise is proportionately enhanced
  - Not really change the performance degradation
- Precoding (Pre-equalization): with fading inverted at the transmitter
  - Requires transmitter knowledge of channel gains
  - No noise enhancement
  - Obtaining accurate and frequent channel estimate difficult for rapidly changing fading channel

# Subcarrier Fading-Compensations

- Coding across subcarriers: Coding with interleaving over time and frequency
  - Bits in a codeword experience different fading
  - Errors associated with few bad subchannels can be corrected
  - Takes advantage of frequency diversity inherent in multicarrier
  - Works best for uncorrelated subcarrier fading (small Bc)
- Adaptive loading: Adapts power and rate on subcarriers relative to their gain
  - Optimization similar to that of adaptive modulation in time (variable rate–variable power MQAM)

# Adaptive Loading

- **Capacity**

$$C = \max_{P_i : \sum P_i = P} \sum_{i=0}^{N-1} B_N \log_2\left(1 + \frac{\alpha_i^2 P_i}{N_0 B_N}\right) \qquad \boxed{\frac{\gamma_i P_i}{P}}$$

$$\frac{P_i}{P} = \begin{cases} 1/\gamma_c - 1/\gamma_i & \gamma_i \geq \gamma_c, \\ 0 & \gamma_i < \gamma_c, \end{cases} \qquad \gamma_i = \alpha_i^2 P / N_0 B_N \qquad C = \sum_{i:\gamma_i \geq \gamma_c} B_N \log_2\left(\frac{\gamma_i}{\gamma_c}\right)$$

- **Adaptive QAM**

$$R = \sum_{i=0}^{N-1} B_N \log_2\left(1 + \frac{K \gamma_i P_i}{P}\right)$$

$$\frac{K P_i}{P} = \begin{cases} 1/\gamma_K - 1/\gamma_i & \gamma_i \geq \gamma_K \\ 0 & \gamma_i < \gamma_K \end{cases} \qquad R = \sum_{i:\gamma_i \geq \gamma_K} B_N \log\left(\frac{\gamma_i}{\gamma_K}\right)$$

# Example

In this problem we investigate multicarrier modulation with adaptive modulation assuming 4 independent AWGN subchannels. Suppose you have a total bandwidth of 100 KHz, a total transmit power $P = 400$ mW, and the noise power in each subchannel is 1 mW. With equal power of 100 mW transmitted on each subchannel, the received SNR on each subchannel is $\gamma_1 = 5$ dB, $\gamma_2 = 5$ dB, $\gamma_3 = 10$ dB, $\gamma_4 = 12$ dB.

(a) [**7 pts**]

What is the capacity of this channel assuming perfect transmit and receive channel side information? How about with just perfect receive side information?

(b) [**6 pts**]

Find the maximum data rate that can be transmitted for a target BER of $10^{-3}$ on each subchannel and the corresponding power allocation policy. Assume each subchannel has unrestricted MQAM modulation and use the BER bound BER $\leq .2e^{-1.5\gamma/(M-1)}$ for your calculations.

(c) [**4 pts**]

Let the average BER be denoted as

$$\bar{P}_b = \frac{1}{N} \sum_{i:\text{Subchannel } i \text{ is used}} P_{b,i},$$

where $P_{b,i}$ is the BER on the $i$th subchannel and $N \leq 4$ is the number of subchannels used i.e. the ones that are allocated non-zero transmit power. If we restrict the answers for the constellations in part (b) to be integer powers of 2 without violating the target BER, find the average BER for your system under the power allocation policy in part (b).

(d) [**8 pts**]

Using the same constellations for each subchannel as found in part (c), can you change the power allocation for each subchannel to lower the value of *average* BER than the value in part (c), under the same total power constraint? If so, find one such power allocation and the corresponding average BER value.

# Solution

(a) Since SNRs are given for $P_i = 100$mW, full-power SNR's are 6 dB higher: $\gamma_1 = \gamma_2 = 11$dB, $\gamma_3 = 16$dB, $\gamma_4 = 18$dB. Next we find $\gamma_0$. Assuming $\gamma_0 < \gamma_i, \forall i$, we have

$$\sum_{i=1}^{4}\left(\frac{1}{\gamma_0} - \frac{1}{\gamma_i}\right) = 1, \Rightarrow \frac{4}{\gamma_0} = 1 + \sum_{i=1}^{4}\frac{1}{\gamma_i}$$

This gives $\gamma_0 = 3.33 < \gamma_i, \forall i$, hence our assumption was correct. Now capacity is

$$C = \sum_{i=1}^{4} B_N \log_2\left(\frac{\gamma_i}{\gamma_0}\right) = \frac{100\text{KHz}}{4}(2 \times 1.917 + 3.578 + 4.243) = 291.4 \text{ Kbps}$$

With CSIR only, equal power allocation is done at the transmitter i.e. $P_i = 100$, so we can just use the $\gamma_i$ values given in the statement of the problem: : $\gamma_1 = \gamma_2 = 5dB$ , $\gamma_3 = 10$dB, $\gamma_4 = 12$dB. So

$$C = \sum_{i=1}^{4} B_N \log_2(1 + \gamma_i) = 291.2 \text{ Kbps}$$

(b) $K = \frac{-1.5}{\ln(5P_b)} = 0.2831$. The SNR's are $\gamma_1 = \gamma_2 = 11$dB, $\gamma_3 = 16$dB, $\gamma_4 = 18$dB. The optimal power allocation strategy is

$$\frac{KP_i}{P} = \begin{cases} \frac{1}{\gamma_K} - \frac{1}{\gamma_i} & \gamma_i \geq \gamma_K \\ 0 & \text{otherwise} \end{cases}$$

with the constraint that $\sum_{i=1}^{4} \frac{P_i}{P} = 1$. This implies

$$\sum_{i=1}^{4} \frac{1}{K} \left( \frac{1}{\gamma_K} - \frac{1}{\gamma_i} \right)_+ = 1$$

Assuming $\gamma_K < \gamma_i, \forall i$, we get $\gamma_K = 8.283 < \gamma_i, \forall i$, hence our assumption was correct. The power allocation policy is $P_1 = P_2 = 58.35$mW, $P_3 = 135.10$mW, $P_4 = 148.20$mW. Now using $M_i = \frac{\gamma_i}{\gamma_K} = 1 + K\gamma_i \frac{P_i}{P}$, we get $M_1 = M_2 = 1.519$, $M_3 = 4.806$, $M_4 = 7.617$. Total data-rate is $R = \sum_{i=1}^{4} \log_2(M_i) = 2 \times 15.078 + 56.621 + 73.231 = 160$ Kbps.

(c) We first find the received SNR's for the 4 channels

$$10\log\gamma_i = db_i \qquad \gamma_i = 10^{0.1 db_i}$$

Received $\quad SNR_i = \dfrac{\gamma_i P_i}{P}$

$$\gamma_1 = \gamma_2 = 10^{1.1}\frac{58.35}{400} = 1.83$$

$$\gamma_3 = 10^{1.6}\frac{135.10}{400} = 13.45$$

$$\gamma_4 = 10^{1.8}\frac{148.20}{400} = 23.37$$

Now, we restrict the constellations to be powers of 2, not exceeding the current value i.e. $M_1 = M_2 = 0$, $M_3 = 4$, $M_4 = 4$. So only the last two subchannels are used. Using the definition,

$$P_{b,3} = 0.2e^{-1.5\frac{\gamma_3}{4-1}} = 2.4 \times 10^{-4}$$

$$P_{b,4} = 0.2e^{-1.5\frac{\gamma_4}{4-1}} = 1.68 \times 10^{-5}$$

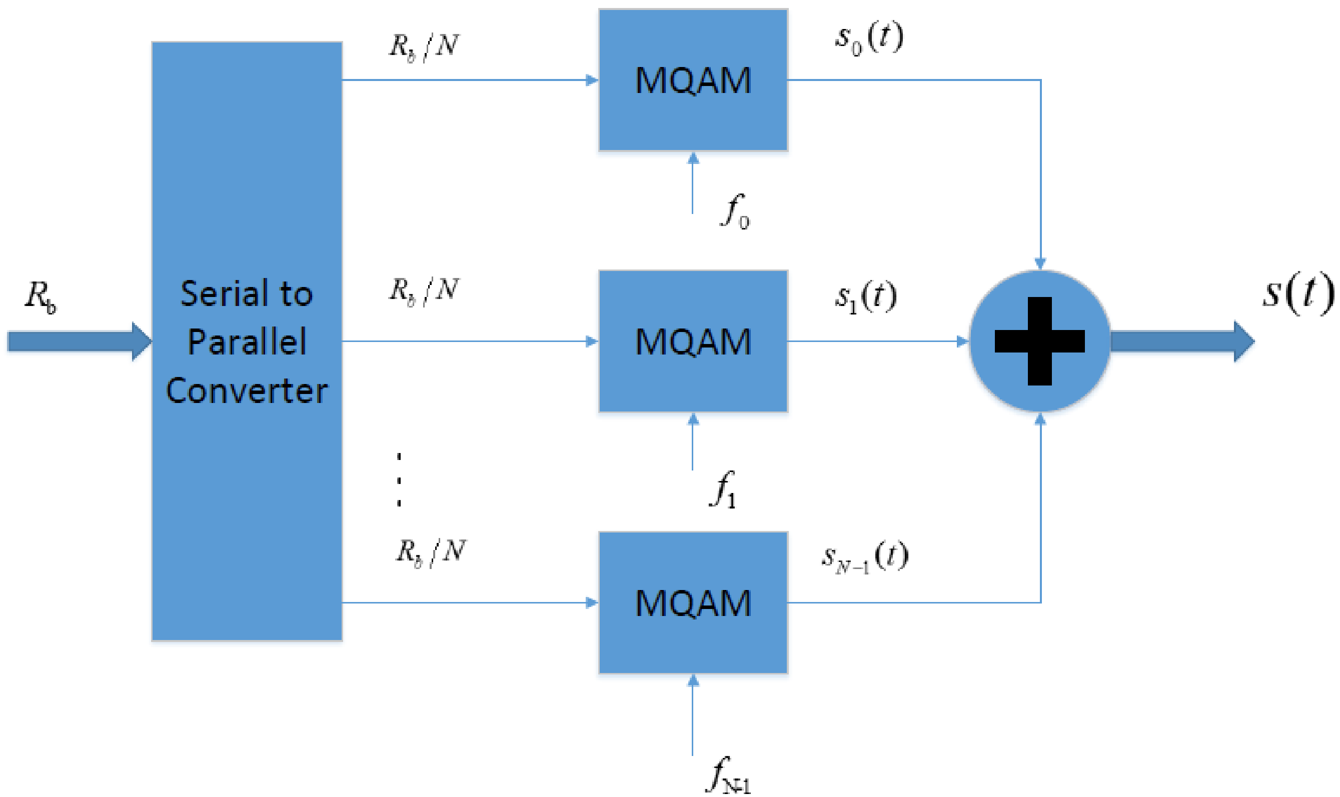
$$\bar{P}_b = \frac{1}{2}(P_{b,3} + P_{b,4}) = 1.21 \times 10^{-4}$$

(d) The constraint is to use $M_1 = M_2 = 0$, $M_3 = 4$, $M_4 = 4$ and find a power allocation s.t. $\bar{P}_b$ is less than $1.21 \times 10^{-4}$. There are many possible solutions, including one which forms this as an optimization problem and solves for the best power allocation. For any power-allocation $P_3$ and $P_4$ satisfying $P_3 + P_4 \leq 400\text{mW}$, we have

$$\bar{P}_b = \frac{1}{2}\left(0.2e^{-1.5\frac{10^{1.6}\frac{P_3}{400}}{3}} + 0.2e^{-1.5\frac{10^{1.8}\frac{P_4}{400}}{3}}\right)$$

The idea is to give more power to the better channel and so essentially do water-filling in some sense. One guess is $P_3 = 155\text{mW}$, $P_4 = 245\text{mW}$, which gives $\bar{P}_b = 9.44 \times 10^{-7} < 1.21 \times 10^{-4}$.

# Multicarrier for QAM Schematic Diagram



# Generation of Multicarrier for QAM

$$s_i(t) = s_{iI}\cos(2\pi f_i t) - s_{iQ}\sin(2\pi f_i t)$$

$$= Re\left\{\left[s_{iI} + js_{iQ}\right]e^{j2\pi f_i t}\right\}$$

$$= Re\left\{s_i e^{j2\pi f_i t}\right\}, f_i = f_0 + i\Delta f = f_0 + \frac{i}{T_N}$$

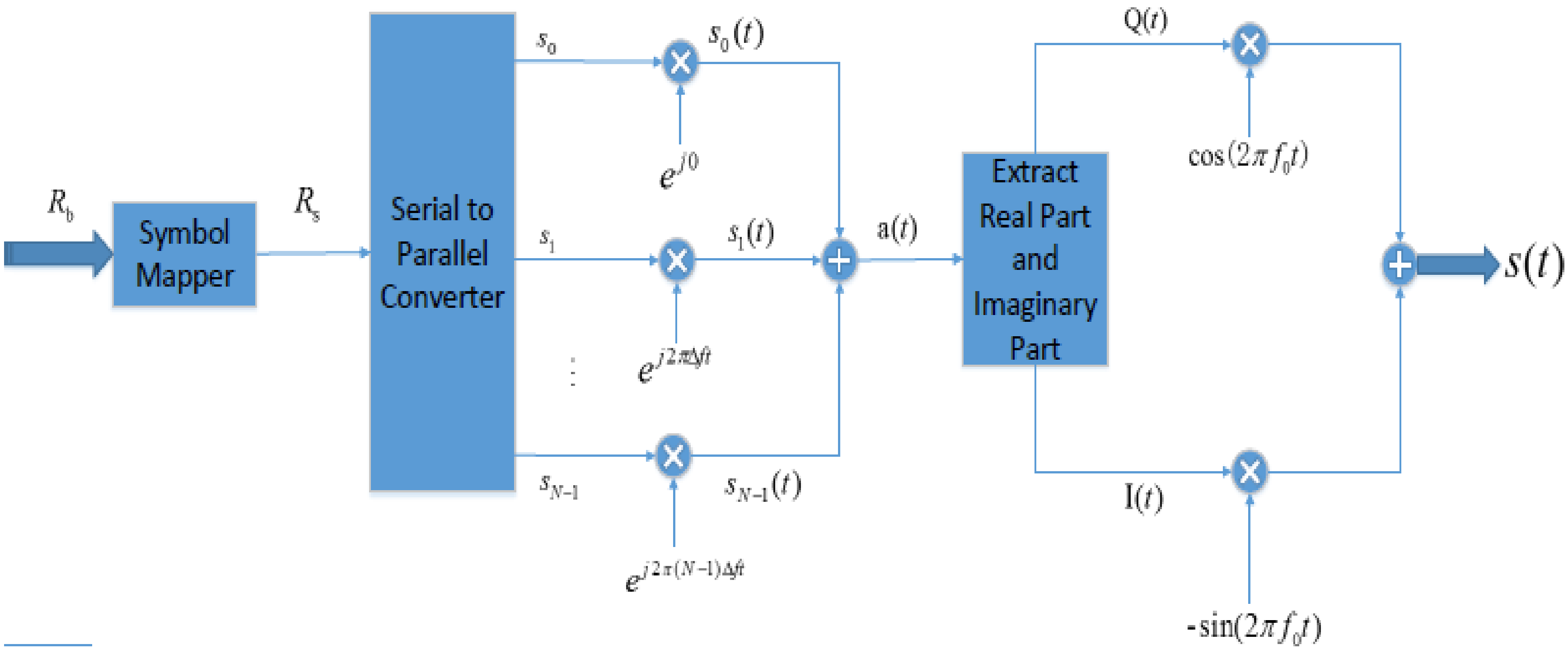
$$s(t) = \sum_{i=0}^{N-1} s_i(t) = Re\left\{\left[\sum_{i=0}^{N-1} s_i e^{j2\pi i\Delta f t}\right]e^{j2\pi f_0 t}\right\}$$

$$= Re\left\{a(t)\, e^{j2\pi f_0 t}\right\}$$

$$a(t) = I(t) + jQ(t) = \sum_{i=0}^{N-1} s_i e^{j2\pi i\Delta f t}$$

$$s(t) = I(t)\cos(2\pi f_0 t) - Q(t)\sin(2\pi f_0 t)$$

# Multicarrier Transmitter for QAM and Rectangular g(t)



# Multicarrier Modulation Discrete Implementation based on DFT

- Implementing N separate modulators and demodulator is very complex and costly
- MCM can be effectively implemented using IFFT at transmitter and FFT at receiver
- The IFFT shifts modulated symbols to desired subcarriers
- A Cyclic Prefix (CP) is inserted in data blocks at transmitter to remove ISI between blocks.  It also makes linear convolution with the channel circular
- Popularity of OFDM is due to the use of IFFT/FFT which have efficient implementations

# Discrete Fourier Transform

Discrete-time sequence: x[n], n = 0, 1, …., N-1

Discrete-frequency sequence: X[i], i = 0, 1, …., N-1

$$X[i] = \frac{1}{\sqrt{N}} \sum_{n=0}^{N-1} x[n] e^{-j2\pi ni/N} , \quad 0 \le i \le N-1$$

$$x[n] = \frac{1}{\sqrt{N}} \sum_{i=0}^{N-1} X[i] e^{j2\pi ni/N} , \quad 0 \le n \le N-1$$

$$\mathrm{x[n]} \underset{\mathrm{IDFT}}{\overset{\mathrm{DFT}}{\rightleftarrows}} \mathrm{X[i]}$$

# Discrete LTI System

**For a Linear Time Invariant channel with discrete input x[n], output y[n], and impulse response h[n]:**

$$y[n] = h[n] * x[n] = x[n] * h[n] = \sum_k h[k]x[n-k]$$

**For circular convolution :**

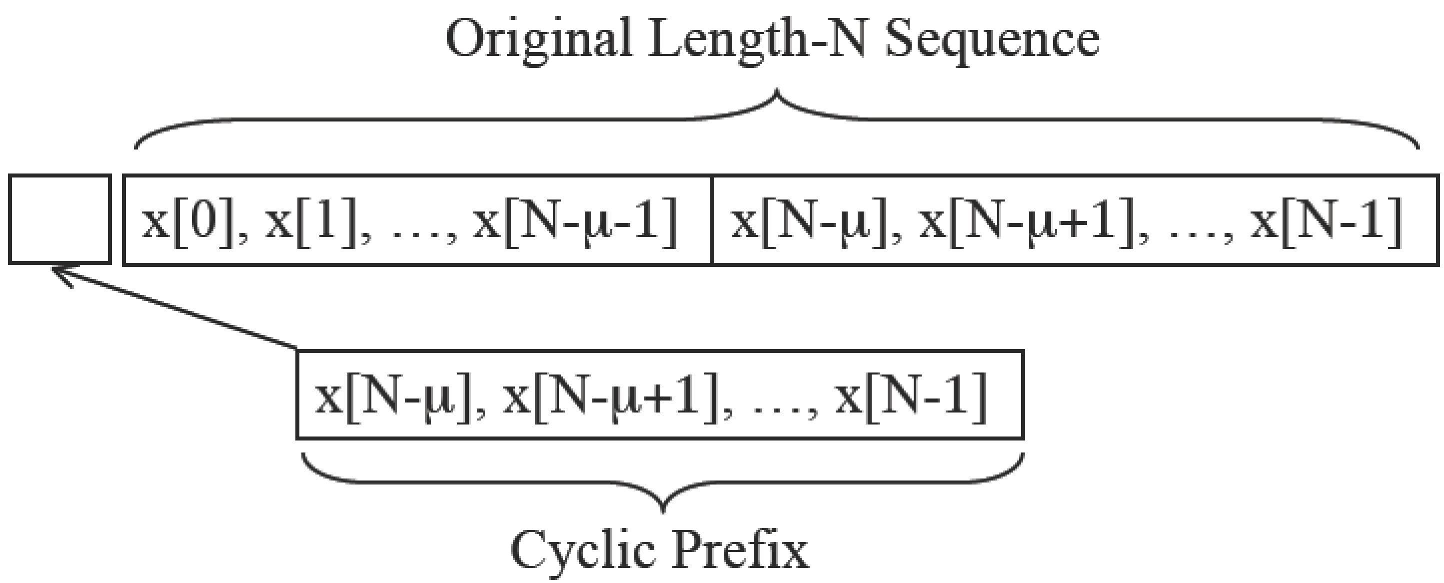
$$y[n] = h[n] \otimes x[n] = x[n] \otimes h[n] = \sum_k h[k]x[n-k]_N$$

$\otimes$ **is circular convolution and $[n\text{-}k]_N$ denotes [n-k] modulo N**

**$Y[i]=X[i]H[i]$, $0 \leq i \leq N-1$**

**$X[i]=DFT[y(n)]/H[i]$**

# The Cyclic Prefix

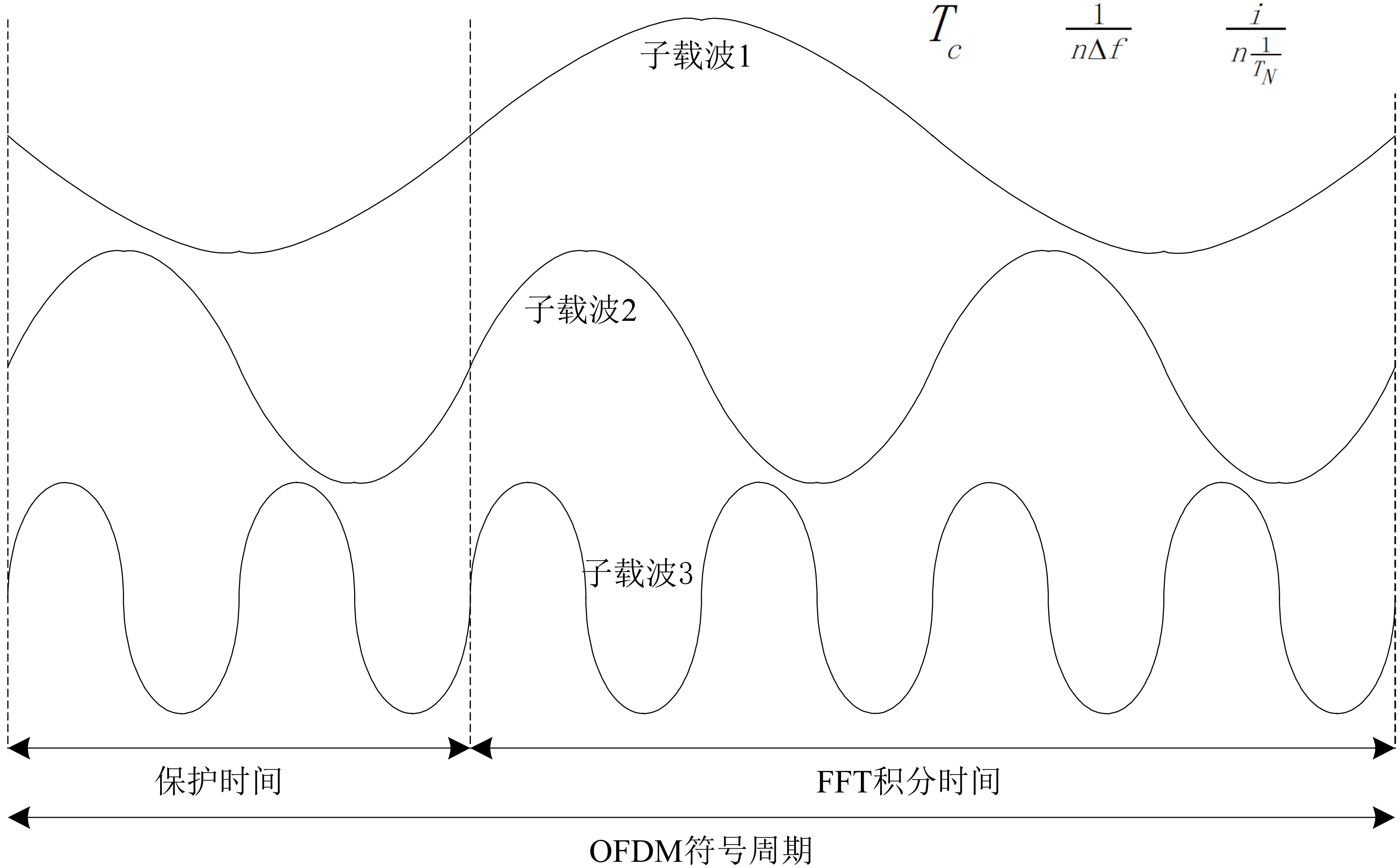



Let $\tilde{x}[n]$ $(-\mu \leq n \leq N-1)$ denote $x[n]$ + cyclic prefix

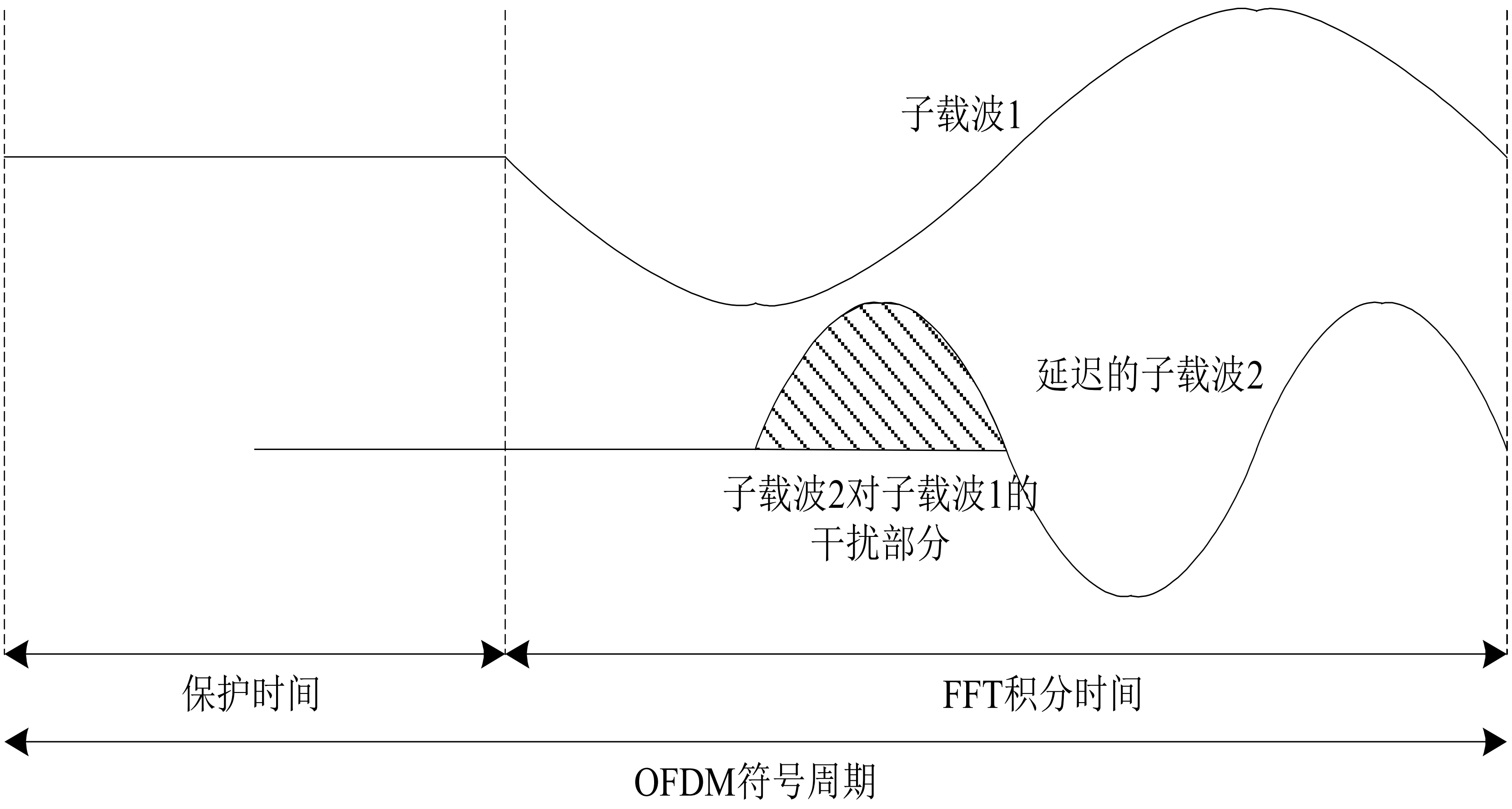
$$\tilde{x}[n] = x[n]_N \longrightarrow y[n] = \tilde{x}[n] * h[n] = x[n] \circledast h[n]$$

# Another Explanation of CP

$$\frac{T_N}{T_c} = \frac{T_N}{\frac{1}{n\Delta f}} = \frac{T_N}{\frac{i}{n\frac{1}{T_N}}} = n$$



# Another Explanation of CP



# Example

考虑一个OFDM系统，系统总带宽B=1MHz，信道时延扩展Tm=5us，每个子信道采用16QAM，发送端信号采用128-IFFT产生，循环前缀（CP）包含8个样值。求：

（1）子载波间隔？

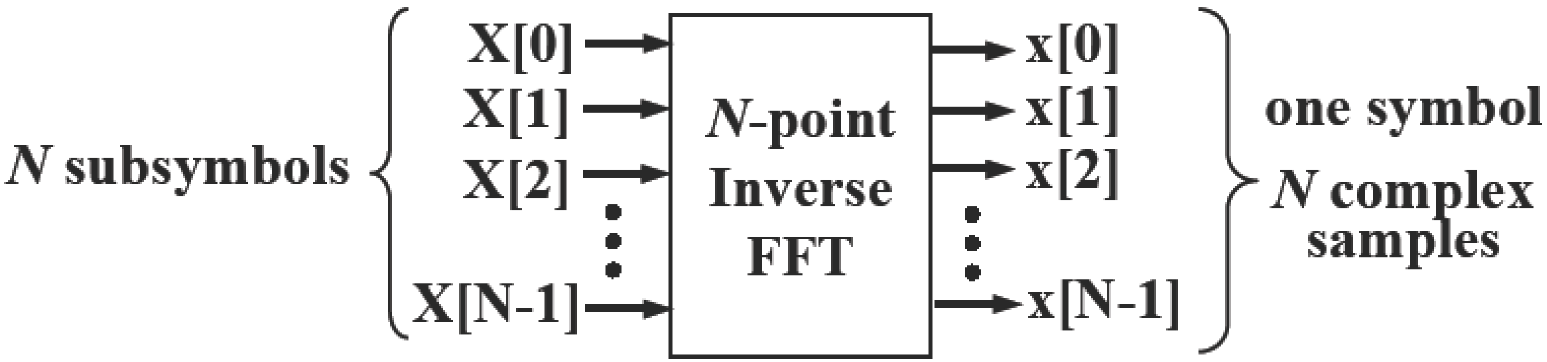
（2）每个OFDM符号的传输时间？

（3）CP的长度能否确保无ISI？

（4）系统的比特率？

（5）如果不加CP，系统的比特速率？

# Orthogonal Frequency Division Multiplexing (OFDM)




$$x[n] = \frac{1}{\sqrt{N}} \sum_{i=0}^{N-1} X[i] e^{j2\pi n i/N}, \quad 0 \le n \le N-1$$

X[i] is a complex symbol stream, output of QAM modulator after passed through a serial-to-parallel converter

x[n] corresponds to samples of the multicarrier signal

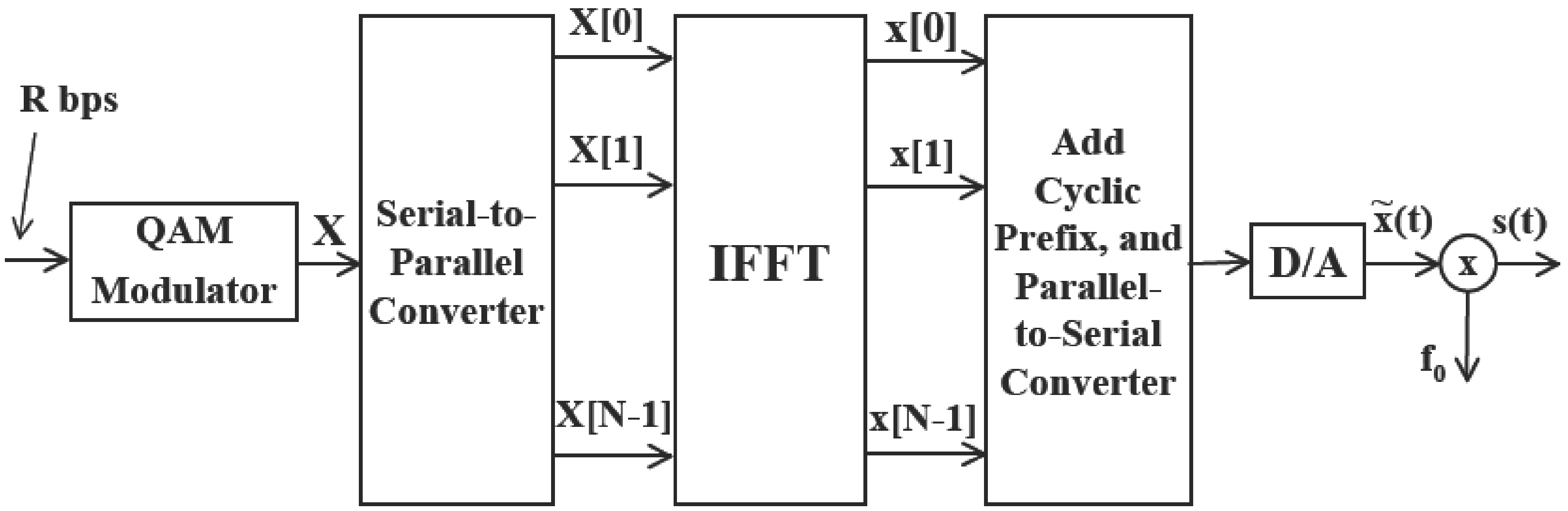
# OFDM with IFFT/FFT

- $x(n)=IFFT[X(i)]$
- Add Cyclic Prefix
- $y(n)=x(n)\otimes h(n)$

  Circular Convolution with $Y(i)=X(i)H(i)$
- $Y(i)=FFT[y(n)]$
- $X(i)=Y(i)/H(i)$

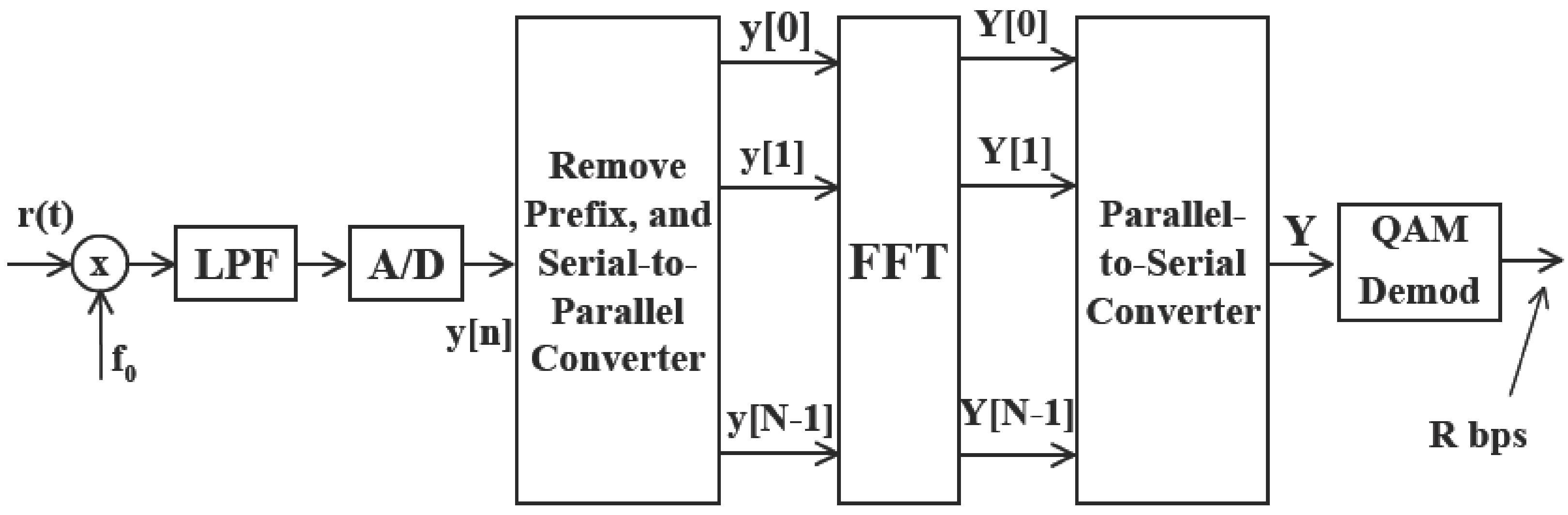
# OFDM with IFFT/FFT

## Transmitter Structure:



# OFDM with IFFT/FFT

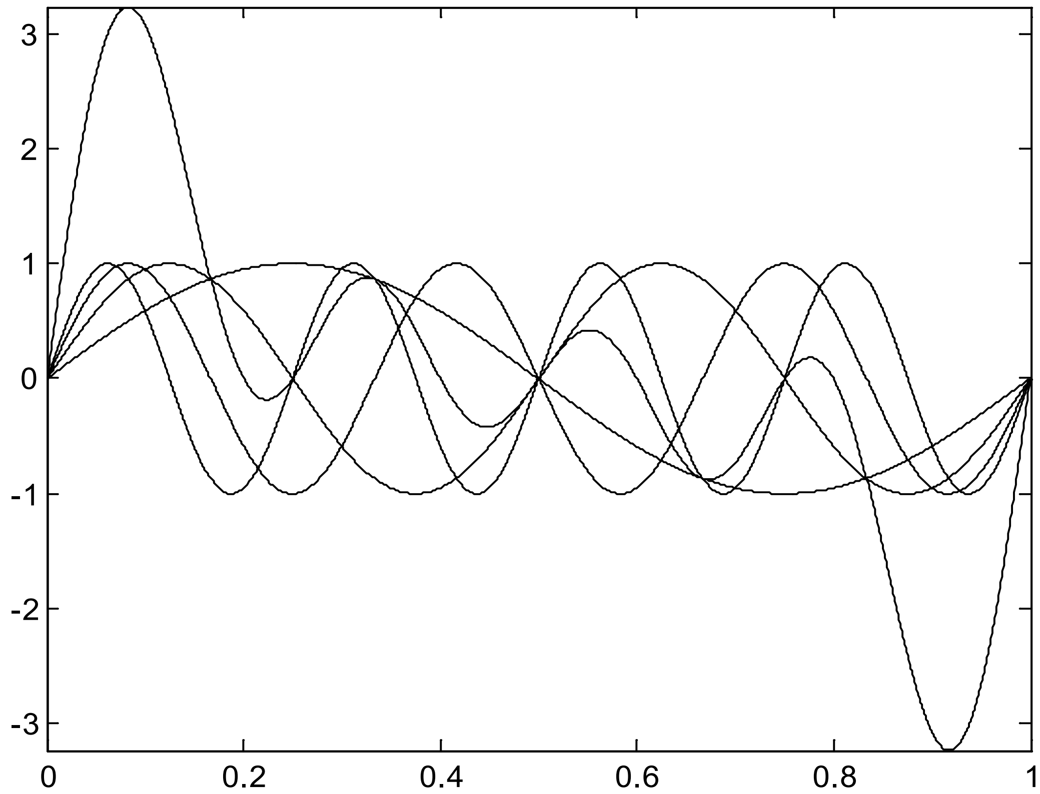
## Receiver Structure:



# OFDM Challenges

- Peak-to-average power ratio
  - Adding multiple substreams can result in high peak signal values
  - Impacts amplifier efficiency
  - Solutions include clipping and coding
- Intercarrier Interference
  - Frequency and timing offset causes interference between carriers
  - Mitigated by minimizing the number of subcarriers and non-rectangular pulse shaping

# Peak to Average Ratio,N=4,data=[1,1,1,1]



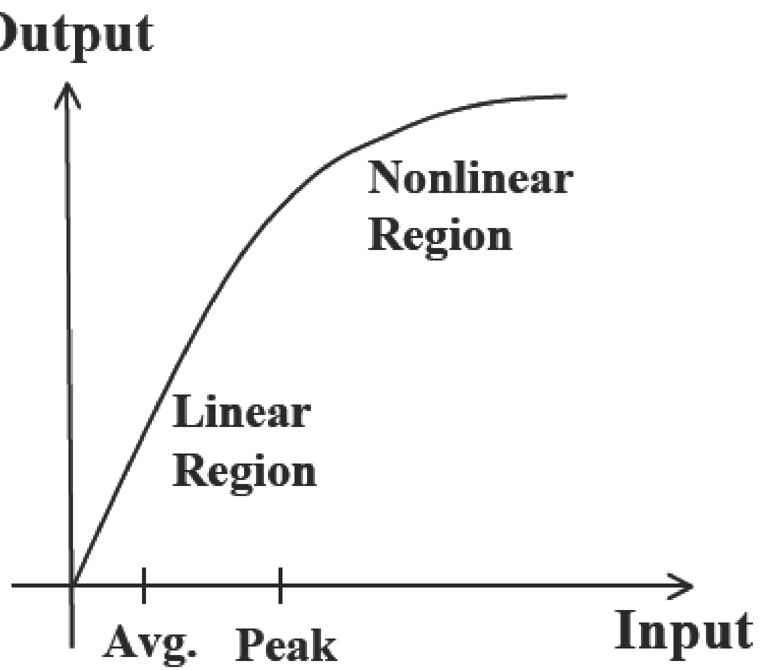
# Peak to Average Ratio

MC signals have high PAR as compared to single carrier systems

PAR increases with the number of subcarriers

$$PAR = \frac{\max_t |x(t)|^2}{\mathbf{E}_t\left[|x(t)|^2\right]}$$

$$PAR = \frac{\max_n |x[n]|^2}{\mathbf{E}_n\left[|x[n]|^2\right]}$$



# Peak to Average Ratio

$$x(t) = A\cos(2\pi f_0 t + \theta)$$

$P_{avg} = A^2/2$, $\quad P_{Peak} = A^2$ $\quad\longrightarrow\quad$ PAR = 2 = 3dB

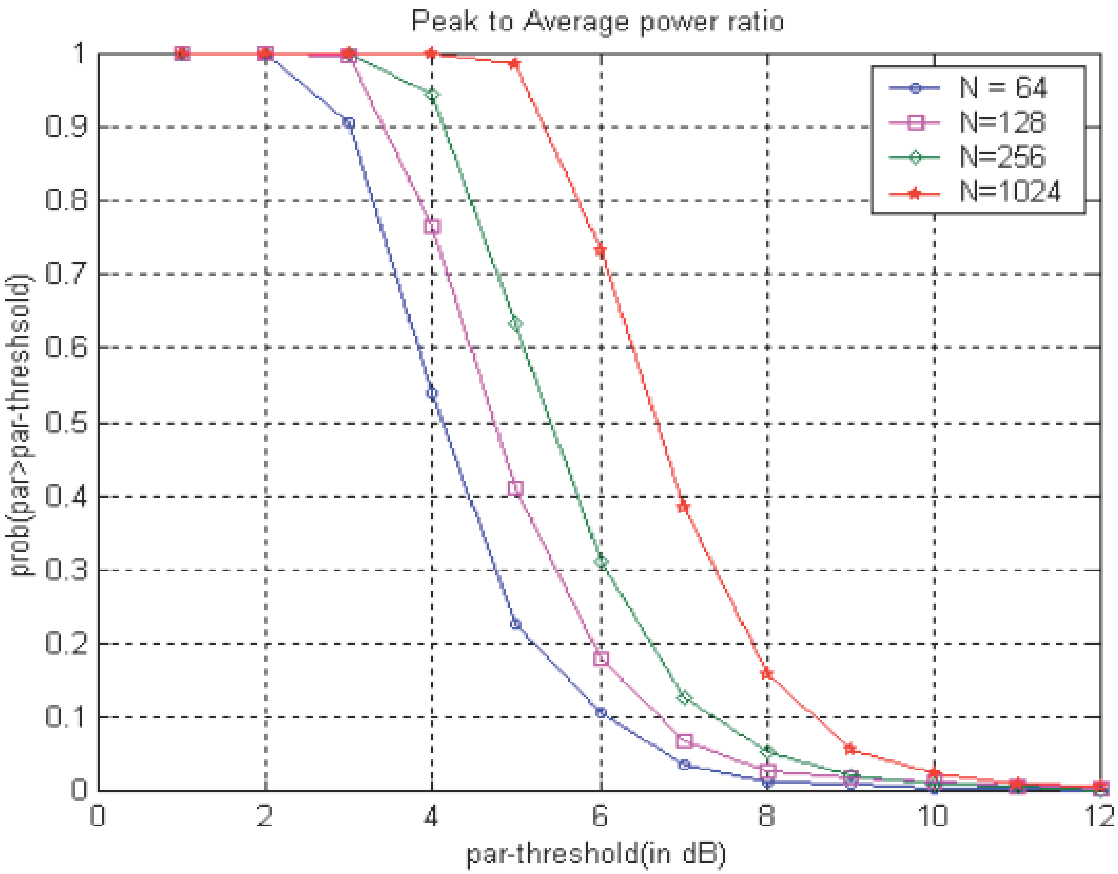
$$x[n] = \frac{1}{\sqrt{N}} \sum_{i=0}^{N-1} X[i] e^{j2\pi in/N}, \qquad 0 \le n \le N-1$$

It can be shown that PAR increases approximately linearly with number of subcarriers N

$$\begin{aligned} E\left[\frac{1}{\sqrt{N}}|x_0 + x_1 + \cdots + x_{N-1}|\right]^2 &= \frac{1}{N} E\,|x_0 + x_1 + \cdots + x_{N-1}|^2 \\ &= \frac{E|x_0^2| + E|x_1^2| + \cdots + E|x_{N-1}^2|}{N} \\ &= 1 \end{aligned}$$

$$\max\left[\frac{1}{\sqrt{N}}\,|x_0 + x_1 + \cdots + x_{N-1}|\right]^2 = \left|\frac{N}{\sqrt{N}}\right|^2 = N.$$

# Peak to Average Ratio



# Intercarrier Interference (ICI)

**The received OFDM signal after radio frequency demodulation:**

$$r(t) = e^{j2\pi\delta_f t}\sum_{i=0}^{N-1}\mathbf{s}_i e^{j2\pi f_i t} g(t)$$

**where** $\delta_f$ **and** $\mathrm{g}(t)$ **represent frequency offset and windowing function, respectively.**

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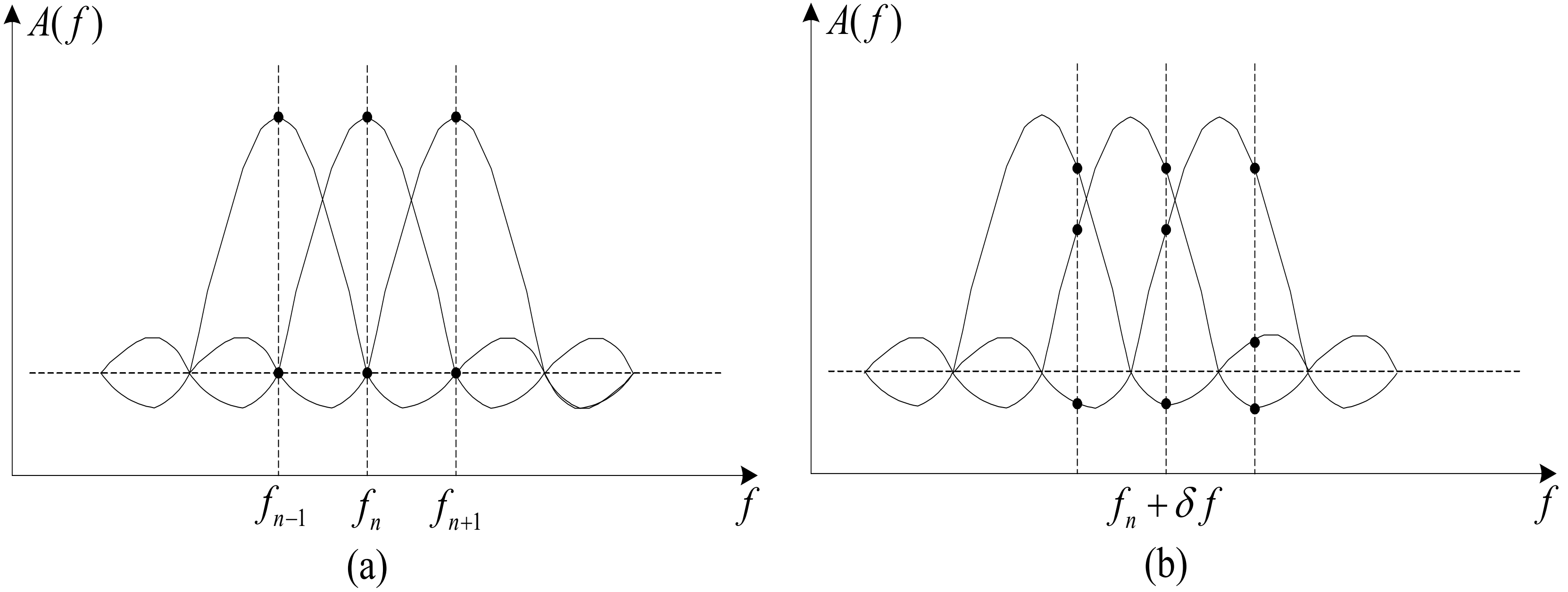
**The decision variable of the nth sub-channel correlation demodulator is**

$$\bar{\mathbf{s}}_n = \int_0^{T_N} r(t) e^{-j2\pi f_n t} dt$$

$$= \mathbf{s}_n \int_0^{T_N} g(t) e^{j2\pi\delta_f t} dt + \sum_{\substack{i=0 \\ i \neq n}}^{N-1} \mathbf{s}_i \int_0^{T_N} g(t) e^{j2\pi(f_i - f_n + \delta_f)t} dt$$

$$= \mathbf{s}_n \, G(-\delta_f) + \sum_{\substack{i=0 \\ i \neq n}}^{N-1} \mathbf{s}_i \, G\left(\frac{n-i}{T_N} - \delta_f\right)$$

# Intercarrier Interference (ICI)



# Intercarrier Interference (ICI)

- **Mismatched oscillator, Doppler shift cause subchannel interference.**
- **Mitigated by minimizing the number of subchannels , synchronization and receive windowing etc.**

# Suppressing ICI—Receive Windowing

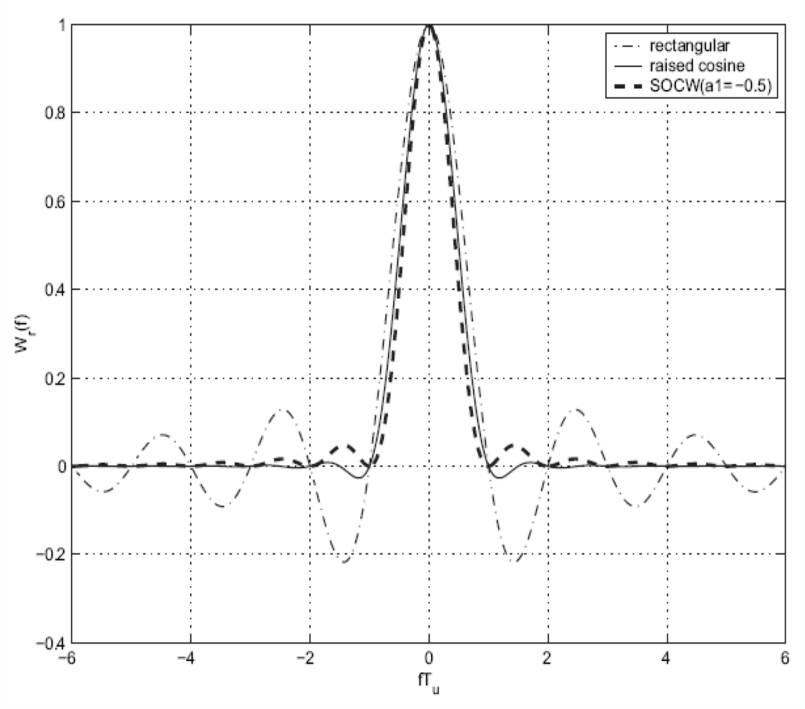



Fig. 2. Frequency spectra for different windows. $\alpha = 1$.

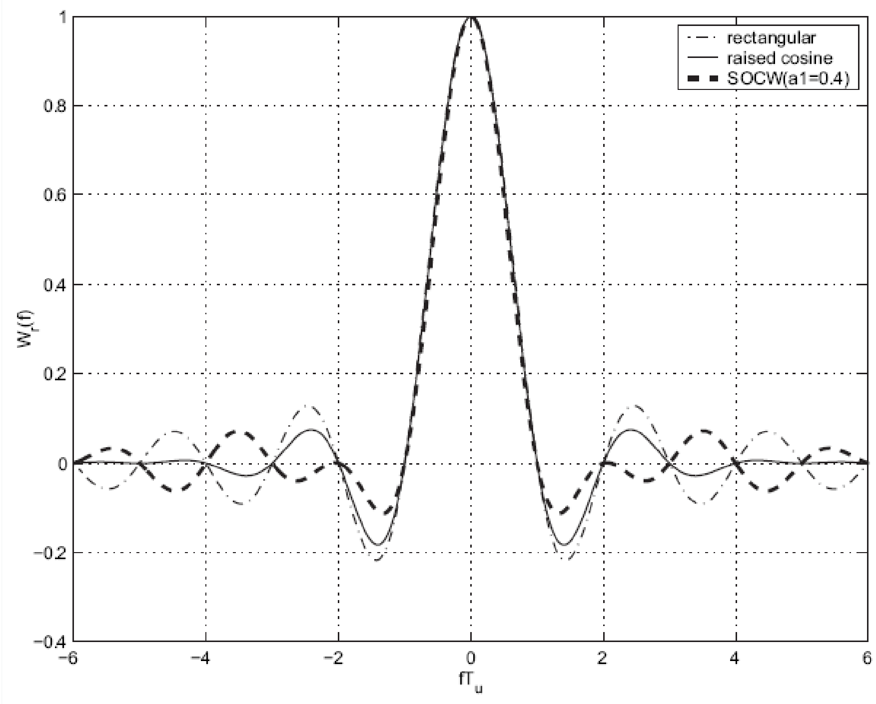



Fig. 3. Frequency spectra for different windows. $\alpha = 0.3$.

# Main Points

- ISI can be mitigated through equalization, multicarrier modulation, or spread spectrum
- Multicarrier is alternative to equalization
  - Splits data stream into NB flat fading substreams
  - Subchannels overlap and still maintain orthogonality
- Fading across subcarriers degrades performance
  - Compensate through frequency equalization, precoding, or adaptive loading
- Discrete implementation of MCM involves use of DFT
  - Cyclic Prefix eliminates ISI and restores orthogonality at RX
- OFDM is efficiently implemented using IFFT/FFT
- OFDM challenges include PAR and timing/frequency offset

- Ex. 12.5, 12.6